

Informatics for Chemistry, Biology, and Biomedical Sciences

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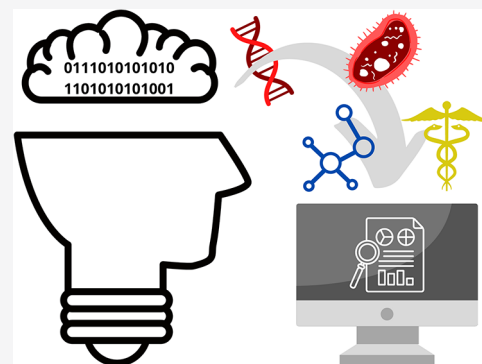


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ABSTRACT: Informatics is growing across disciplines, impacting several areas of chemistry, biology, and biomedical sciences. Besides the well-established bioinformatics discipline, other informatics-based interdisciplinary fields have been evolving over time, such as chemoinformatics and biomedical informatics. Other related research areas such as pharmacoinformatics, food informatics, epi-informatics, materials informatics, and neuroinformatics have emerged more recently and continue to develop as independent subdisciplines. The goals and impacts of each of these disciplines have typically been separately reviewed in the literature. Hence, it remains challenging to identify commonalities and key differences. Herein, we discuss in context three major informatics disciplines in the natural and life sciences including bioinformatics, chemoinformatics, and biomedical informatics and briefly comment on related subdisciplines. We focus the discussion on the definitions, historical background, actual impact, main similarities, and differences and evaluate the dissemination and teaching of bioinformatics, chemoinformatics, and biomedical informatics.



INTRODUCTION

Oxford's dictionary¹ defines *informatics* as “the study of processes for storing and obtaining information”. Cambridge's dictionary² defines informatics as a discipline that “studies the structure, behavior, and interactions of natural and artificial systems that store, process, and communicate information”. Without doubt, computers have played a significant role in storing, processing, and disseminating information. Accordingly, Oxford's dictionary defines information as “data as processed, stored, or transmitted by a computer” in computing. As such, the critical importance of computers to rapidly transform data or facts into information and information into knowledge through inductive learning has been steadily increasing.³

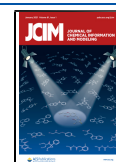
Chemistry, biology, and biomedical sciences rely on data collected in numerous ways, for instance, through experimental observations and measurements, medical observations, or computer simulations, to name a few. Advances in technologies such as high-throughput synthesis and testing, automated, robotic miniaturized sequencing, proteomics, glycomics, lipidomics, and other “-omics” approaches are examples of the main drivers of informatics concerning the massive volumes of data that are generated. With the advent of “big data” across many fields, the amount of information and its complexity are growing exponentially, accompanied by increasing demands for advancements in informatics in terms of computational power and information networking infrastructure to precisely store, process, disseminate, model, analyze, and predict such information. For instance, from 2006 to 2018, data related to investigational drugs available in

the DrugBank database increased by almost 300%, drug–drug interactions data by nearly 600%, and the number of single nucleotide polymorphisms (SNP)-associated drug effects increased by more than 3000%.⁴ Close relationships between chemistry, biology, and medicine (and more confined or specialized areas such as food chemistry, material, and polymer sciences, or neurobiology) with informatics have prompted the development of specific research disciplines that interface informatics with other fields. Relationships between such research fields are illustrated schematically in Figure 1. For our discussion, we organize the major research fields associated with informatics into three predominant areas: bioinformatics, chemoinformatics, and biomedical informatics. There are other informatics-related subdisciplines evolving around the three main fields (Figure 1) that are also discussed in the following sections. Additional relationships could be established that would further extend and complicate the picture presented in Figure 1.

Historically, bioinformatics evolved as an informatics discipline in the life sciences during the 1960s as a natural consequence of using informatics in healthcare that already began in the 1950s (*vide infra*). In addition, chemical information (science) and chemoinformatics (cheminfor-

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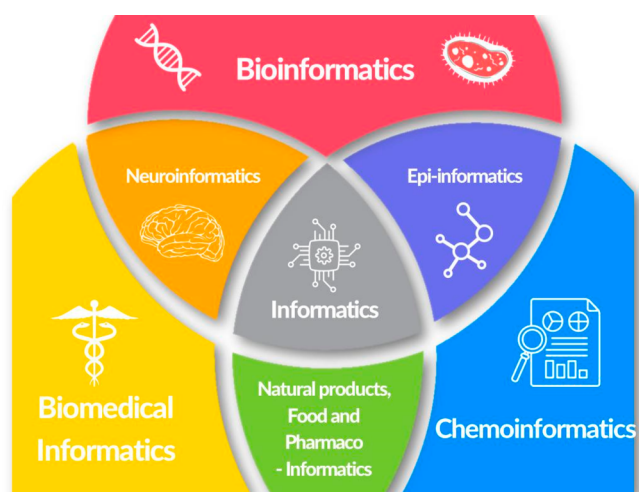


Figure 1. Relationships between chemoinformatics, bioinformatics, biomedical informatics, and related subdisciplines.

matics) also originated during the 1960s, long before these terms were coined, but evolved on a smaller scale. Bioinformatics developed alongside biomedical informatics, which have shared roots. There are other currently evolving disciplines associated with informatics, for example, materials informatics, polymer informatics, food informatics, and additional subdisciplines such as epi-informatics (Figure 2) that might be rationalized as subdisciplines of the three major fields. The boundaries between these fields, as illustrated in Figure 1, are rather fluid and not rigidly defined. This is due to the strong interdisciplinary areas of these disciplines and their ongoing development. For example, in drug discovery, chemoinformatics, bioinformatics, and biomedical informatics meet and often share similar informatics approaches. It is a characteristic feature of these fields that their methodological foundations are partly overlapping, yet distinct.⁵ Thus, we also find terms in the literature such as bio/cheminformatics, biocheminformatics, or chem-bioinformatics that reflect these intrinsic relationships.⁶ Notably, while it is very common to find bioinformatics courses in academic curricula at the

undergraduate or graduate level, chemoinformatics education currently is still rare.

The following discussion is primarily focused on the bioinformatics, chemoinformatics, and biomedical informatics disciplines. Exemplary formal definitions proposed in the literature and a brief history are presented. We also specify common concepts and goals as well as distinguishing differences and comment on other subdisciplines and informatics-related research fields that are just beginning to emerge. It is hoped that this discussion will aid in providing a clearer picture of the strong influence of informatics on chemistry and the life and medical sciences. This contribution also aims to support the correct use of terminology among educators and students. Of note, we intentionally do not specifically center the discussion on artificial intelligence or deep learning, given the many recent contributions on these topics.^{3,7–9}

■ INFORMATICS-RELATED RESEARCH DISCIPLINES

Impact of Informatics on Chemistry and Biology.

Over time, the number of scientific publications in the above-mentioned informatics-related disciplines has significantly increased, as illustrated in Figure 3, which reflects the evolution of these fields. Today, it becomes nearly impossible to develop further chemistry and biology without the use of computers and specialized software tools. One of the main reasons for the increasing use of computational concepts in experimental files is that increasing amounts of data and information must be made accessible in databases and retrieved for research. Informatics is now essential to store, process, analyze, and disseminate heterogeneous data in biology, chemistry, and beyond. Moreover, algorithms and data structures from informatics are essential for deriving predictive models that often complement and guide experimental programs. Predictions of chemical and biological properties and simulations of molecular systems have become integral components of basic and applied research. In chemistry, for example, planning and designing of synthetic reactions has added a new dimension to organic synthesis; in biology, numerical simulations of physiological processes and their dynamics have further extended insights from experiments and yielded

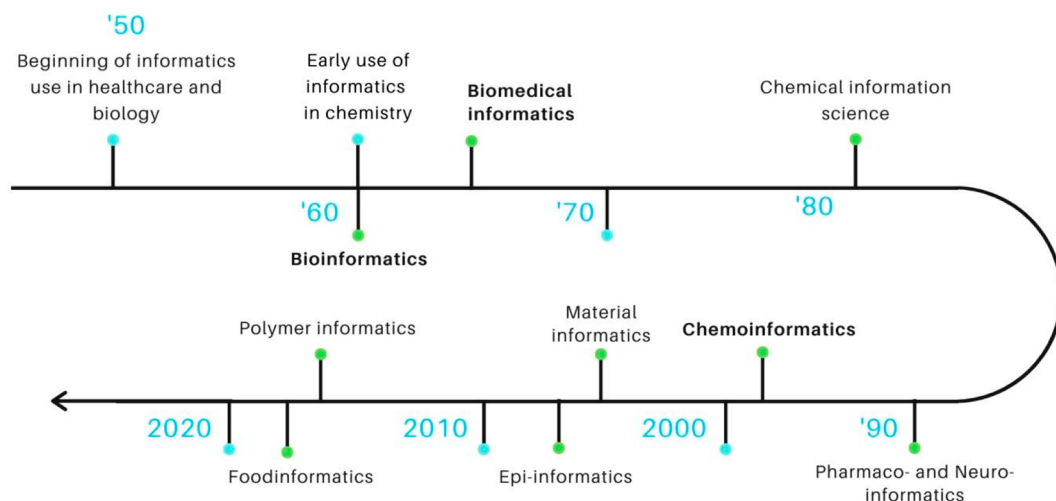


Figure 2. Informatics in the literature. The term of each discipline is marked on a time scale when it appeared for the first time, as searched in PubMed and SciFinder[®].

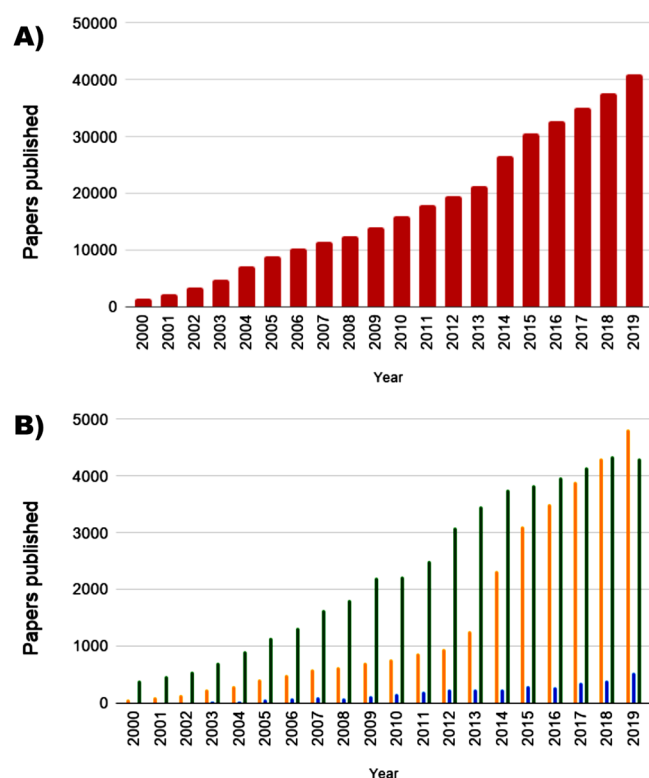


Figure 3. Growth of informatics-related disciplines. Reported is the number of references for informatics-related disciplines in the literature including (A) publications with the keyword “bioinformatics” (red) and (B) publications with keywords “chemoinformatics” or “cheminformatics” (blue), “biomedical informatics” (orange), and “materials informatics” (green) in PubMed.

new testable hypothesis. Furthermore, in both fields, computational visualization methods have become indispensable for representing and characterizing chemical and biological spaces of increasing complexity such as those constituted by ultralarge compound arrays or expression profiles. Moreover, without image analysis approaches from informatics, modern high content screening campaigns, phenotypic analysis, or image-based medical diagnostics would not be feasible. As a consequence, as discussed further in the next sections, the demand for trained experts on bioinformatics and chemoinformatics is steadily increasing. In particular, finding scientists trained in bioinformatics working in hospitals, universities, public or private research institutions, and companies is common by now. Similarly, the demand for scientists with expertise in informatics is increasing in the chemical sciences in both academia and industry, although education in chemoinformatics still operates on a smaller scale, despite continuous growth in recent years.

Definitions. For the evolving informatics-related disciplines and subdisciplines, various formal definitions have been proposed, as summarized in Table 1. Additional definitions from the literature for some of these disciplines are further discussed below.

Scientific Dissemination and Education. The development of these informatics-oriented disciplines has also triggered the introduction and consolidation of peer-reviewed scientific journals accounting for each area. Table 2 summarizes examples of peer-reviewed journals that include the word “informatics” and related terms such as “information”

Table 1. Exemplary Definitions of Research Associated with Informatics and Associated Terms

Term	Definitions published in the literature ^a	ref
Bioinformatics	“Bioinformatics is a research field in which computer scientists, biologists, physicians, mathematicians, and chemists combine their expertise to collaborate on diverse tasks, from the discovery of new facts on complex biological systems to the rationalization of organization of health systems.”	10
Chemoinformatics/cheminformatics, chemical informatics ^b	“Provides computer methods for learning from chemical data and for modeling tasks a chemist is facing”. “All concepts and methods that are designed to interface theoretical and experimental efforts involving small molecules”. “Chemoinformatics is a generic term that encompasses the design, creation, organization, management, retrieval, analysis, dissemination, visualization, and use of chemical information.”	5, 11, 12
Materials informatics	“It is a rapidly developing field of research which is engaged with the application of informatics principles to analyze materials-related data.”	13
Polymer informatics	“Polymers data-driven machine learning studies”.	14
Pharmaco informatics	“It is the scientific field that utilizes a systemic approach to process and utilize medication-related data and information, including its acquisition, storage, analysis, and dissemination in the delivery of optimal medication-related patient care and health outcomes.”	15
Food informatics	“It is the use of chemical information methodologies to address food-related challenges.”	16
Biomedical informatics	“It is the application of the science of information as data plus meaning to problems of biomedical interest.”	17
Neuroinformatics	It is the application of “advanced informatics methods to deal with the flood of neuroscientific data; by developing and applying data analysis methods for the study of the brain; by providing both analytical and numerical tools for theoretically modeling brain function; and by exploiting our insights into the principles underlying brain function”.	18
Epi-informatics	It is the “discovery and development of small molecule epigenetic drugs and probes features multidisciplinary strategies with strong computational approaches that have led to the successful discovery and/or optimization of compounds that act as modulators of epigenetic targets”.	19

^aTextual definition as published in the reference cited. Additional definitions are in the main text. ^bAlternative names frequently found in the literature.

Table 2. Examples of Peer-Reviewed Journals in the Biological, Chemical, and Biomedical Sciences Containing the Word Informatics or Information in the Journal's Title^a

Journal	Publisher	Publication period	Journal URL
<i>Bioinformatics</i>	Oxford Academic	1985–present	https://academic.oup.com/bioinformatics
<i>BMC Bioinformatics</i>	BMC-Springer Nature	2000–present	https://bmcbioinformatics.biomedcentral.com/
<i>Briefings in Bioinformatics</i>	Oxford Academic	2000–present	https://academic.oup.com/bib
<i>Genomics, Proteomics & Bioinformatics</i>	Elsevier	2003–present	https://www.journals.elsevier.com/genomics-proteomics-and-bioinformatics/
<i>Molecular Informatics</i>	Wiley	2010–present	https://onlinelibrary.wiley.com/journal/18681751
<i>Journal of Cheminformatics</i>	BMC-Springer Nature	2009–present	https://jcheminf.biomedcentral.com/
<i>Journal of Chemical Information and Computer Sciences</i>	American Chemical Society	1975–2004	–
<i>Journal of Chemical Information and Modeling</i>	American Chemical Society	2005–present	https://pubs.acs.org/loi/jcisid8
<i>Chemical Information Science Gateway</i>	F1000Research	2015–present	https://f1000research.com/cis/
<i>Journal of the American Medical Informatics Association (JAMIA)</i>	Oxford Academic	1994–present	https://academic.oup.com/jamia
<i>IEEE Journal of Biomedical and Health Informatics</i>	IEEE Engineering in Medicine and Biology Society	1991–present	https://www.embs.org/jbhi/
<i>International Journal of Medical Informatics</i>	Elsevier	1997–present	https://www.journals.elsevier.com/international-journal-of-medical-informatics
<i>Journal of Biomedical Informatics</i>	Elsevier	2001–present	https://www.journals.elsevier.com/journal-of-biomedical-informatics
<i>Frontiers in Neuroinformatics</i>	Frontiers	2007–present	https://www.frontiersin.org/journals/neuroinformatics

^aTable lists exemplary journals. The number of specialized journals in the areas of bioinformatics, chemoinformatics, and biomedical informatics is reported in Table 3.

in the title. Of note, there are many more computational journals in chemistry, biology, and biomedical sciences (in addition to multidisciplinary journals that also publish studies focusing on chemoinformatics, bioinformatics, and biomedical informatics). The representative journals in Table 2, together with the publication period and the publisher, further emphasize the development and consolidation of these individual disciplines. While scientific journals and textbooks reflect the evolution of these fields, so do formal courses in educational programs at the undergraduate and graduate levels that have been established, as summarized in Table 3. Of note, Table 3 includes the source information for the numbers provided and aims to show that the number of bioinformatics journals, textbooks, and courses outnumber those available for chemoinformatics and biomedical informatics. This is at least

Table 3. Current State of Teaching and Dissemination of Three Main Informatics-Related Disciplines

Discipline	Bioinformatics	Chemoinformatics	Biomedical informatics
Undergraduate programs ^a	~500	~30	~30
Master programs ^b	~450	~10	~120
Textbooks ^{cd}	364/~300	17/~100	22/~80
Specialized journals ^e	25	3	2
Proceedings of international conferences ^f	117	3	42

^aResults retrieved from <https://www.bachelorsportal.com/> (accessed October, 2020). ^bResults retrieved from <https://www.mastersportal.com/> (accessed October, 2020). ^cResults retrieved from SciFinder^h: <https://scifinder-n.cas.org/> (accessed November, 2020). ^dResults retrieved from Google Books: <https://books.google.com/> (accessed November, 2020). The numbers in this table account for books that contain the keywords “bioinformatics”, “chemoinformatics,” or “biomedical informatics” in the title of the book. ^eResults retrieved from <https://www.scimagojr.com/> (accessed October, 2020).

in part due to the fact that medical curricula have as of yet limited opportunities to flexibly expand into theory. Furthermore, chemistry departments are traditionally conservative in implementing informatics education. In fact, much of the chemoinformatics discipline, as we perceive it today, has originated from the pharmaceutical industry rather than academic settings, which also explains the strong orientations of chemoinformatics toward drug discovery.

Bioinformatics. The history and scope of bioinformatics have been reviewed in detail.^{20,21} Briefly, the 1950s and 1960s (Figure 2) encompass the birth of computational biology with many developments that shaped bioinformatics as an increasingly independent discipline. For example, Margaret Dayhoff is a pioneer in bioinformatics and has been considered “the mother and father” of this discipline.²¹ The term “bioinformatics” itself was coined and defined by Paulien Hogeweg and Ben Hesper in 1970, which conceptualized it as “the study of informatic processes in biotic systems”.²² With time, the conceptual framework of bioinformatics has further evolved, and more general definitions have been proposed. An exemplary definition is provided in Table 1. Alternative definitions include “the computational handling and processing of genetic information” and “the application of computational methods to the field of biochemistry”.²⁰ The reader is also referred to previous reviews for a detailed account of the history and milestones of bioinformatics development.^{20,21} Bioinformatics has matured to process biological information on a large scale at the interface with experimental biology.

Typical areas of study in bioinformatics can be grouped into four broadly defined areas including sequence analysis, molecular database development, protein structure prediction, and study of molecular evolution. Accordingly, other informatics-driven subdisciplines of bioinformatics (not containing the “informatics” term) include systems biology, synthetic biology, or whole-cell modeling.²⁰ Arguably, neuroinformatics and epi-informatics can also be regarded as subdisciplines of bioinformatics (Figure 1).

The importance and continued application of informatics to biological systems catalyzed the introduction of peer-reviewed journals that included advances in bioinformatics. As early as 1961 commenced publication of the *Journal of Theoretical Biology* that continues to exist to this date. Other early journals that published informatics contributions to solve biological systems problems included *Computers in Biology and Medicine* (established in 1970) and *Computational Biology and Chemistry* (1976). Table 2 summarizes other peer-reviewed journals that contain “bioinformatics” in the journal’s name. These journals were selected from the more than 25 peer-reviewed journals related to bioinformatics that are currently published (Table 3 and Supporting Information). As shown in Table 2, some of these current journals date back to 1985, especially *Bioinformatics*, a mainstay in this field. The journal *Molecular Informatics* (Table 2) positions itself at the interface between bioinformatics and chemoinformatics (e.g., it uses the term “bio/cheminformatics” in the journal’s description). In contrast to chemoinformatics, bioinformatics is a general indexing term in specialized scientific journals.

Recently, it was commented on the need to define the bioinformatician profession more clearly.²² To this end, universities are integrating bioinformatics as an independent discipline into the curricula of biology, medicine, and related fields (Table 3). More than 360 textbooks have been published on bioinformatics to support teaching and learning (Table 3 and Supporting Information).

Chemoinformatics. The history of chemoinformatics has been extensively discussed in several excellent reviews.^{5,23–25} The origin of informatics in chemistry date back to the late 1950s to early 1960s, addressing problems related to structure representation, structure elucidation, synthesis design, and chemical data collection, analysis, and management, among several others, with subsequent and significant applications to drug discovery. However, the term chemoinformatics itself was first coined and defined by Brown and Bristol in 1998: “The use of information technology and management has become a critical part of the drug discovery process. Chemoinformatics is the mixing of those information resources to transform data into information and information into knowledge for the intended purpose of making better decisions faster in the area of drug lead identification and organization.”²⁶ Additional definitions have been put forward.^{26,27} Two of them are in Table 1. The emergence of chemoinformatics in the historical context of other informatics-related disciplines and subdisciplines is shown in Figure 2. For a more in-depth discussion of the history of this discipline, the reader is referred to previous publications with the historical roots of chemoinformatics.^{5,23–25}

Chemoinformatics shares commonalities with theoretical chemistry and computational chemistry such as quantum chemistry and force field-based molecular modeling. However, there are significant conceptual differences as discussed by Varnek and Baskin.²⁸ For instance, the use of distinct molecular representations constitutes one of the major differences between chemoinformatics and areas related to theoretical chemistry. Another major difference is the inductive learning (inference) mechanism, which plays a dominant role in chemoinformatics. It has been argued that chemoinformatics and bioinformatics are subdivisions of biomedical informatics.²⁹ However, chemoinformatics has been established as an independent discipline with a strong focus on drug discovery.¹¹

Currently, informatics approaches are applied to different areas in chemistry, leading to the emergence of new subdisciplines (Figure 1, Table 1). For example, “food informatics,” (or “food chemical information”) was originally introduced in 2014 and encompasses the application of informatics to food chemicals.^{30,31} Notably, the topic of informatics in food science has been included in a recently published textbook of chemoinformatics.³² Other nascent subdisciplines of chemoinformatics include “polymer informatics”³³ and “materials informatics”,³⁴ (Table 1). Despite the emergence of these more specialized subdisciplines, chemoinformatics continues to be broadly applied across several areas of chemistry.³³

The development of chemoinformatics has also been accompanied by the introduction of peer-reviewed journals by different publishers (Table 2). The first of these journals was the *Journal of Chemical Information and Computer Sciences*, introduced in 1975, which succeeded the *Journal of Chemical Documentation* initially established in 1961. The *Journal of Chemical Information and Computer Sciences* became a cornerstone for the development of the chemoinformatics discipline and was then divided in 2005 into the *Journal of Chemical Information and Modeling*, which was positioned as a successor for chemoinformatics research and molecular modeling, and the *Journal of Chemical Theory and Computation*, which was oriented toward theoretical/computational chemistry. Today, there are several journals with chemical information or related terms in the title (Table 1). Examples include the *Journal of Cheminformatics* (established in 2009) and *Chemical Information Science Gateway* from F1000Research (2015), which represents the first open review publication platform for chemoinformatics. Other journals such as *Molecular Informatics* (that succeeded *QSAR and Combinatorial Science* in 2010) publish papers in the fields of both chemoinformatics and bioinformatics (*vide supra*). Even though the number of scientific journals dedicated to chemoinformatics is smaller than the number of bioinformatics journals (Table 3), these publication venues are of critical importance for the further development of chemoinformatics as an independent discipline. However, chemoinformatics is yet to be added as an indexing term to specialized as well as multidisciplinary journals.

In addition to the introduction of scientific journals, the increasing importance of this discipline (and increasing demand for chemoinformatics scientists in the industry) has prompted the need to create textbooks, courses, and university programs at the undergraduate and graduate levels to aid in formal training. As stated above, the number of textbooks and courses on bioinformatics outnumber similar resources to teach chemoinformatics (Table 3). However, since the first publication of the *Handbook of Cheminformatics* by Gasteiger in 2003,³⁵ the number of specialized textbooks and edited books for teaching and disseminating chemoinformatics information is increasing. The Supporting Information includes a list of representative textbooks, tutorials, and other teaching-oriented books for chemoinformatics. In 2020, the *Journal of Cheminformatics* (Table 2) implemented a new type of research article with the designation “educational”. This article type aims to introduce chemoinformatics to newcomers to the field ranging from undergraduate students to experienced chemists primarily focused on experimental subfields of chemistry.³⁶ Wild and Wiggins have reviewed the academic programs and specializations offered at academic institutions and concluded

Table 4. Comparison of Concepts and Techniques Distinguishing Informatics-Based Disciplines

	Bioinformatics	Chemoinformatics	Biomedical informatics
Main objective and scope; type of data	Macromolecules and genes	Small molecules and property annotations	Health-related data knowledge
Core identity, data representation, or query	- Sequences - Structures (coordinates)	- Molecular fingerprints - Numerical descriptors	Clinical/nonclinical patterns
Estimated data size available (data volume)	Petabytes of information (10^{15} bytes) ⁵³	Terabytes of information (10^{12} bytes) ⁵⁰	Petabytes of information (10^{15}) ⁵⁴
Sources of data (drivers)	"Omics"-technologies - Array/expression data - Robotic, miniaturized sequencing	- High-throughput synthesis and testing - Enumeration of chemical libraries	Signal processing and pattern recognition
Examples of study subfields	- Sequence analysis - Biomolecular databases - Protein structure comparison and prediction - Molecular evolution	- Chemical libraries - Diversity analysis - Property prediction (including biological activity) - Structure generation	- Health databases - Prediction and prognosis of clinical diagnostic - Pharmacovigilance
Key concepts and techniques	- Homology - Alignment	- Chemical space - Similarity - Complementarity	Image and documentary analysis
Examples of overlapping concepts	- Databases: storage and processing - Data mining - Data training - Data visualization - Prediction of properties or outcomes - Protein–ligand/small molecule interactions (at the interface between bioinformatics and chemoinformatics).		

that the number is still limited.³⁷ Formal courses on chemoinformatics are not widely available at present. Similar to "bioinformaticians",¹⁸ there is no well-defined profile for training "chemoinformaticians." However, the increasing demand for specialized chemoinformatics researchers should encourage academic institutions to modify and adapt their chemistry curricula and introduce chemoinformatics or chemical information science course and programs, either as part of chemical training or as a specialization at the masters level.

Biomedical Informatics. In response to increasing data volumes in medicine and the need to process clinical information more systematically, the concept of "biomedical informatics" was introduced. The first implementations were devised by Homer Warner in the 1950s. However, the term was formally coined between the 1960s and 1970s (Figure 2).³⁸ Followed by the increase in storage and knowledge management capacity, this new concept gradually began to play a leading role in decision making related to the design of new study programs, administrative efforts, communication of information, and basic research in medicine. Some reviews enable the assessment and impact of biomedical informatics on the medical community.¹⁶

One of the principal differences between the biomedical informatics discipline and others discussed herein is the broad spectrum of data, information, and knowledge that originates from clinical settings. This discipline is responsible for generating knowledge from clinical and nonclinical data related to health, for example, patient management, continuing education of staff, monitoring, efficacy, safety of treatments,

technical, and human resources.³⁹ Biomedical informatics also has a close knowledge relationship to the "-omic" areas⁴⁰ and bioinformatics, which impacts prognosis, diagnosis, and personalized therapy.

Recently, clinical subdisciplines have emerged such as "clinical informatics" or "nurse informatics".⁴¹ In addition, other areas such as "pharmacovigilance" have increasingly utilized computational concepts to further their development.⁴²

Also, due to the increasing access to computer equipment and modernization of hospital environments (for example, the digitization of clinical records), the biomedical informatics discipline has grown considerably. This is reflected by the creation of new alternatives for teaching and knowledge dissemination in this area. For example, there currently are at least 120 master programs and 22 textbooks, as well as specialized journals, that played a pioneering role for this discipline such as the *IEEE Journal of Biomedical and Health Informatics* (established in 1991) or the *International Journal of Medical Informatics* (1997). The Supporting Information includes a list of representative textbooks, international conferences, and other biomedical informatics resources (shown in Tables 2 and 3 of the Supporting Information).

Other Emerging Informatics-Related Disciplines. In addition to the emerging subdisciplines discussed above, other multidisciplinary areas in life sciences are strongly influenced by informatics. For example, natural product research in drug discovery has been impacted by informatics applications at the interface between chemoinformatics and biomedical informatics (Figure 1). Several recent studies have analyzed the use

of computers to store, organize, and disseminate information related to natural products research and botanics.^{43–46} Also, several recent research papers and reviews highlight the impact of informatics on natural product-based drug discovery.^{29,47–49}

Thus, we envision the emergence of a new field that might be termed “natural products informatics”. Arguably, this new field might further evolve at the interface between chemoinformatics and bioinformatics since the computational analysis of biogenesis is also of significant importance in natural product research beyond drug discovery.

Outside of chemistry and the life sciences, the primary focal points of our discussion, “materials informatics” (Table 2), currently also evolves as a discipline, as mirrored by increasing numbers of publications (Figure 3).

Similarities and Differences between Chemoinformatics and Bioinformatics. Another important question is which might be key similarities and distinguishing features of chemoinformatics and bioinformatics. Table 4 presents a side-by-side comparison of the three main disciplines discussed herein, also including biomedical informatics. The table compares the main objectives and scope of each discipline, approaches to data representation, size of data samples in subdisciplines, key concepts, and techniques. The table also lists representative examples of concepts common to these disciplines.

At the core of the different informatics-driven disciplines is the type of data to be processed and analyzed, for example, macromolecules, peptide sequences, genes (bioinformatics), chemical structures, chemical reactions (chemoinformatics), clinical and physiological data (biomedical informatics). The different types of data require the development of special representations designed to handle, process, and analyze domain-specific data. For example, in bioinformatics, it is common to use sequences and structural coordinates to represent biological macromolecules. In chemoinformatics, it is common to calculate molecular fingerprints and numerical descriptors.

The complexity and size of the data and objects to be represented by each discipline and their subdisciplines translate into partly shared and partly unique challenges for organizing, archiving, processing, and analyzing data and for knowledge extraction. For instance, polymers and nanomaterials are difficult to represent computationally. In food and natural product chemistry, food matrices and mixtures of natural products are also challenging to represent. In drug discovery, metallo-drugs are often disregarded in chemoinformatics because of the complexity to accurately calculate drug-like properties that would enable, for instance, virtual screening of organometallic compounds. Challenges associated with handling domain-specific data due to differences in their dimensionality and size are also considerably different, as indicated in Table 4. Bioinformatics handles much larger amounts of information (on the order of pentabytes) than chemoinformatics (terabytes).⁵⁰ For biomedical informatics, the estimation is more challenging since most of the clinical data is not accessible in the public domain. The explosion of biological data over the past few years has prompted the need to use ontologies in bioinformatics in order to address semantic and organizational differences between biological databases.⁵¹ On the other hand, chemoinformatics faces the challenge to handle exceedingly large numbers of small molecules that populate, for instance, drug-like chemical space,⁵² and manage the associated structure–property

relationships. In bioinformatics, examples of major sources of information are -omics technologies, while in chemoinformatics a major source of data is high-throughput synthesis and testing, as well as the enumeration of on-demand or virtual libraries. By contrast, in biomedical informatics, a major source of information is signal processing.

Furthermore, each discipline addresses unique questions in light of specific needs. For example, as summarized in Table 4, bioinformatics is concerned with sequence analysis, building and maintaining biomolecular databases, conducting protein construction comparison, and predicting facets of molecular evolution. In comparison, chemoinformatics is focused on developing chemical libraries of small molecules, diversity analysis, property prediction including prediction of biological properties, and structure generation. Bioinformatics relies more on cellular data, while chemoinformatics builds upon on *in vitro* data, especially in drug discovery applications. In sharp contrast, biomedical informatics is focused on analyzing health-related data at the clinical level.

The different informatics fields, however, also share problems and challenges, for instance, the ability to store, organize, and manage different types of data. There is no doubt that an increasingly large amount of increasingly diverse and complex data and information needs to be processed to extract new knowledge. Thus, a factor common to all disciplines is the need to handle and process “big data” (biological, chemical, and medical). With the parallel advancements in informatics, the quantity and quality of computational tools and resources (commercial and open-source) are increasing.

Different informatics disciplines can also share methodologies. This is discussed in the recent comprehensive review *QSAR without Borders*.⁵⁵ This work highlights that methods initially developed for establishing quantitative structure–activity relationships (QSAR) in medicinal chemistry and physical organic chemistry have been expanded to nanotechnology, material science, biomaterials, synthesis planning, and clinical informatics. By analogy, it would be possible to arrive at the generalization, “informatics without borders”, at least for certain tasks. For example, similar to QSAR, the concept of an “activity landscape”⁵⁶ originally emerged for drug discovery applications but has been generalized to “property landscapes” including applications to other areas of chemistry^{57–59} and potentially biology.⁶⁰

As stated above, a common objective of all informatics-related disciplines is to convert information into knowledge. Many real-world problems require multidisciplinary approaches and collaboration of two or more research fields. Prominent examples are drug discovery and development projects that frequently involve bioinformatics, biomedical informatics, and chemoinformatics methods.⁶¹ For instance, modeling protein–ligand/small molecule interactions requires predicting the three-dimensional arrangement of the molecular target (a task of the primary interest for bioinformatics) and the management and processing of small molecules, typical of drug-like molecules (a task of primary interest for chemoinformatics). Along the same lines, structure-based virtual screening of medium- to large-sized chemical libraries, molecular dynamics, and homology modeling of biological targets to guide structure-based design, among others methods, are common examples of computer-aided drug design that often require the combination of bioinformatics and chemoinformatics approaches. Hence, despite their distinguishing features, there is substantial potential to further explore and

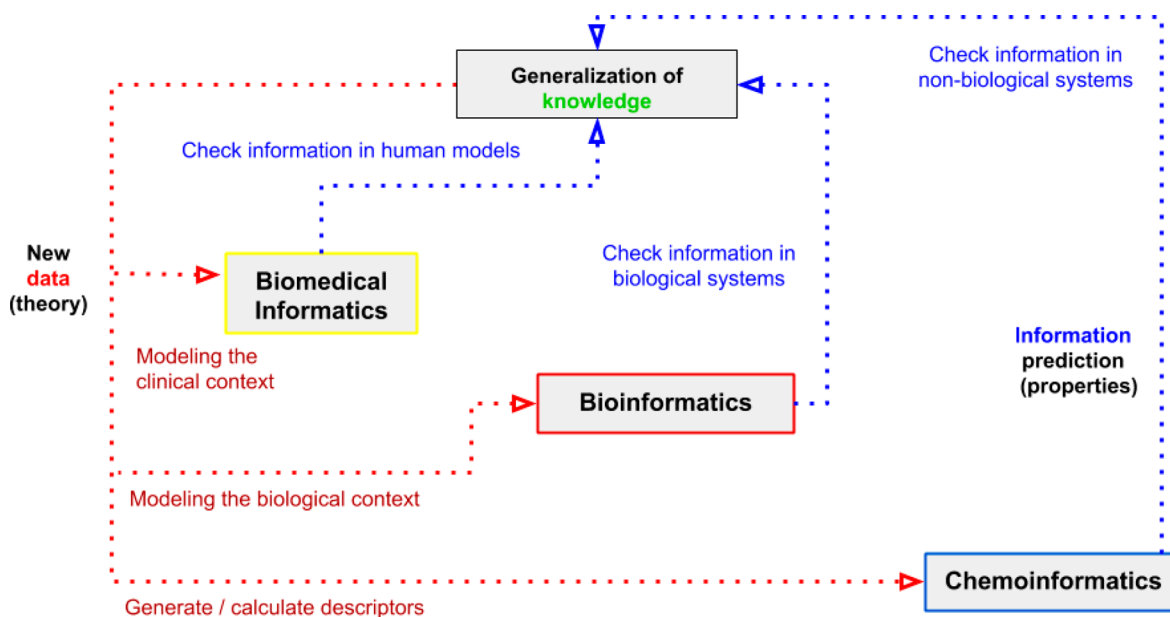


Figure 4. Interrelations between chemoinformatics, bioinformatics, and biomedical informatics. The scheme is the author's point on areas of collaboration between the three major informatics-related life sciences disciplines and the interchange of data, information, and knowledge between them.

exploit the overlap between bioinformatics and chemoinformatics.

Figure 4 depicts a proposal for interactions between chemoinformatics, bioinformatics, and biomedical informatics as part of an inductive–deductive learning cycle. The figure also illustrates the transformation of data into information and, eventually, knowledge. It also shows feedback of the informatics-related disciplines from existing and emerging areas. Given the overlap between these disciplines, their boundaries are fluid and dynamic in nature. However, the further development of each discipline also requires focusing on distinguishing features. To this end, scientific dissemination, teaching, and formal training in well-organized or planned courses would delineate the differences between and common elements of informatics-related disciplines and subdisciplines.

Taken together, the increasing need for experts in chemoinformatics and biomedical informatics and the overall limited number of teaching resources in chemoinformatics strongly suggests the inception of new teaching concepts and programs.

CONCLUSIONS

Herein, we analyzed the impact of informatics across biology, chemistry, and medicine, leading to the emergence of new informatics-based disciplines. In particular, we have emphasized three main disciplines including bioinformatics, chemoinformatics, and biomedical informatics that have conceptual and methodological similarities and distinguishing features. Such disciplines initially emerged and further evolved to solve problems in biology, chemistry, and biomedicine. However, the multidisciplinary nature of complex problems in the life sciences including drug discovery often requires adapting and applying different informatics approaches. In some instances, further specializations (or subdisciplines) have also emerged to address complex and challenging problems such as the case of neuroinformatics at the interface of bioinformatics and biomedical informatics.

A common conceptual overlap between bioinformatics, cheminformatics, and biomedical informatics is that they are driven by data and mostly inductive learning mechanisms. Also, all disciplines rely on databases and face the challenges of data collection, standardization, integration, mining, visualization, and property prediction.

Key distinguishing features of each informatics-related discipline include major goals and study topics. For example, a major distinguishing characteristic is the type of *data representation* used; while bioinformatics employs sequences and coordinates of three-dimensional structures, cheminformatics frequently relies on molecular fingerprints and numerical descriptors. By contrast, biomedical informatics uses clinical or nonclinical patterns encoded in varying formats.

For formal training of scientists, it is convenient to delineate these informatics-related disciplines with soft boundaries and plan and develop educational programs with well-defined objectives. Clearly, attempting to define an overarching “informatics for all” discipline encompassing chemistry, biology, and medicine would be overly broad and take away from the most relevant interesting domain-specific research and teaching topics.

As discussed above, each informatics-related discipline analyzes different types of data using unique representations. We anticipate that in the future, as these disciplines further evolve, various subdisciplines will continue to emerge focusing on more specialized data (e.g., food chemicals, polymers, or natural products). However, since informatics-related disciplines are often applied simultaneously to solve multidisciplinary problems, it is also anticipated that they will continue to operate on a continuous spectrum of data and questions that also requires coordination and partial integration of these efforts. For instance, incorporating cheminformatics into the larger spectrum of chemical information science expands its primary focus on small molecules to chemical data and chemical information from any source.⁶² Similarly, concerning the training and specialization of scientists and practitioners in different informatics-

related disciplines is now increasingly covered under the interdisciplinary umbrella of “data science”. Clearly, striving for a balance between integration and cultivation of discipline-specific features is one of the challenges for informatics-based disciplines going forward.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.0c01301>.

Table S1: List of books of discipline-type informatics.

Table S2: List of specialized journals and international conferences of discipline-type informatics. (PDF)

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Notes

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